Verification & Validation Plan

1. Verification Objectives

- Prove the scheduler actually runs.
- Prove tracked tasks are persisted durably.
- Prove overdue/completion/failure notifications are emitted under the right conditions.
- Prove duplicate notifications are suppressed appropriately.

2. Verification Methods

Area	Method	Evidence
Registry writes	File inspection	JSON contents, timestamps
Checker execution	Scheduler logs	cron/systemd logs, timestamps
Status transitions	Controlled task fixtures	before/after registry state
Notifications	OpenClaw message evidence	sent-message log, target session/chat
Failure handling	Simulated failed process/session	failure update and resolved state

3. Validation Scenarios

1. Task starts and finishes before next scheduled reminder.
2. Task remains active long enough to trigger one or more progress reminders.
3. Task fails and sends an immediate failure update.
4. Task becomes unobservable and emits an honest degraded-status update.
5. System restarts and still remembers active tracked tasks.

4. Proof Package Expectations

For implementation phases, proof should include:

- commands run
- state file snapshots
- scheduler/service logs
- message routing evidence
- before/after task state
- test results for non-trivial logic

5. Validation Exit Criteria

- At least one completion case proven
- At least one overdue reminder case proven
- At least one failure case proven
- At least one restart/persistence case proven
- No false completion claim in any tested scenario